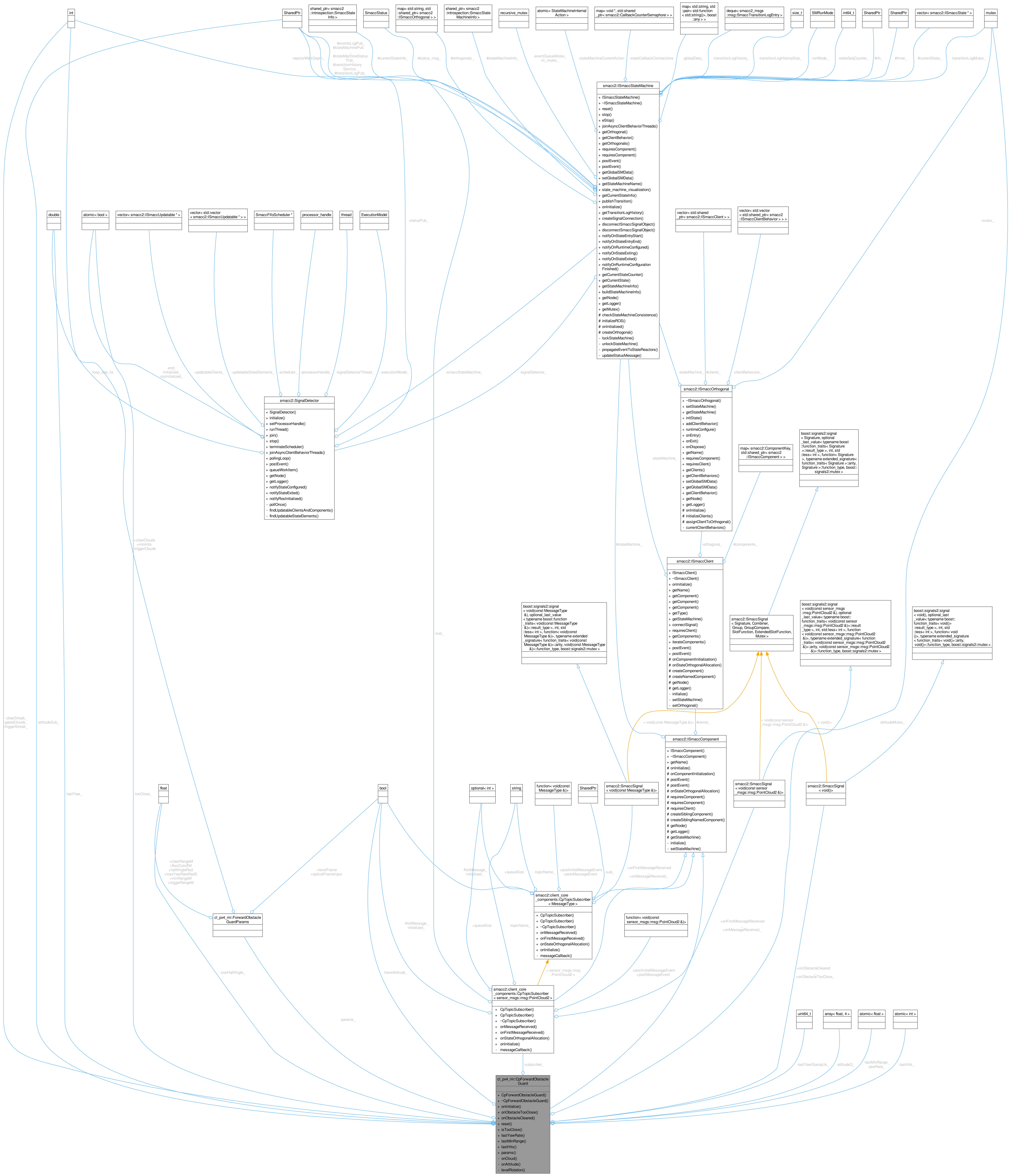
int
SharedPtr
shared_ptr< smacc2::introspection::SmaccStateInfo >
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
shared_ptr< smacc2::introspection::SmaccStateMachineInfo >
recursive_mutex
atomic< StateMachineInternalAction >
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std::pair< std::function< std::string()>, boost::any > >
deque< smacc2_msgs::msg::SmaccTransitionLogEntry >
size_t
SMRunMode
int64_t
SharedPtr
SharedPtr
vector< smacc2::ISmaccState * >
mutex
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfigurationFinished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
-reactorWaitDepth_
#eventsLogPub_
#stateMachinePub_
#stateMachineStatusPub_
#transitionHistoryService_
#transitionLogPub_
#currentStateInfo_
#status_msg_
#orthogonals_
#stateMachineInfo_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-transitionLogHistorySize_
-runMode_
-stateSeqCounter_
#nh_
#timer_
#currentState_
-transitionLogMutex_
double
atomic< bool >
vector< smacc2::ISmaccUpdatable * >
vector< std::vector< smacc2::ISmaccUpdatable * > >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
vector< std::shared_ptr< smacc2::ISmaccClient > >
vector< std::vector< std::shared_ptr< smacc2::ISmaccClientBehavior > > >
-statusPub_
-mutex_
-loop_rate_hz
-end_
-initialized_
-rosInitialized_
-updatableClients_
-updatableStateElements_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-smaccStateMachine_
-signalDetector_
-stateMachine_
#clients_
-clientBehaviors_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
smacc2::ISmaccOrthogonal
+ ~ISmaccOrthogonal()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
map< smacc2::ComponentKey, std::shared_ptr< smacc2::ISmaccComponent > >
boost::signals2::signal< Signature, optional_last_value< typename boost::function_traits< Signature >::result_type >, int, std::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost::signals2::mutex >
-stateMachine_
-orthogonal_
#components_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
#stateMachine_
-clearClouds
+minHits
+triggerClouds
boost::signals2::signal< void(const MessageType &), optional_last_value< typename boost::function_traits< void(const MessageType &)>::result_type >, int, std::less< int >, function< void(const MessageType &)>, typename extended_signature< function_traits< void(const MessageType &)>::arity, void(const MessageType &)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal< Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal< void(const sensor_msgs::msg::PointCloud2 &), optional_last_value< typename boost::function_traits< void(const sensor_msgs::msg::PointCloud2 &)>::result_type >, int, std::less< int >, function< void(const sensor_msgs::msg::PointCloud2 &)>, typename extended_signature< function_traits< void(const sensor_msgs::msg::PointCloud2 &)>::arity, void(const sensor_msgs::msg::PointCloud2 &)>::function_type, boost::signals2::mutex >
boost::signals2::signal< void(), optional_last_value< typename boost::function_traits< void()>::result_type >, int, std::less< int >, function< void()>, typename extended_signature< function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
< void(const MessageType &)>
#owner_
< void(const sensor_msgs::msg::PointCloud2 &)>
< void()>
-attitudeMutex_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
-clearStreak_
-gatedClouds_
-triggerStreak_
-attitudeSub_
-lastYaw_
-tooClose_
float
bool
optional< int >
string
function< void(const MessageType &)>
SharedPtr
smacc2::SmaccSignal< void(const MessageType &)>
smacc2::SmaccSignal< void(const sensor_msgs::msg::PointCloud2 &)>
smacc2::SmaccSignal< void()>
+clearRangeM
+floorCutoffM
+halfAngleRad
+maxYawRateRadS
+minRangeM
+triggerRangeM
+levelFrame
+opticalFrameInput
-firstMessage_
-initialized_
+queueSize
-topicName_
+postInitialMessageEvent
+postMessageEvent
-sub_
+onFirstMessageReceived
+onMessageReceived_
cl_px4_mr::ForwardObstacleGuardParams
smacc2::client_core_components::CpTopicSubscriber< MessageType >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
function< void(const sensor_msgs::msg::PointCloud2 &)>
-cosHalfAngle_
-firstMessage_
-initialized_
-haveAttitude_
+queueSize
-topicName_
< sensor_msgs::msg::PointCloud2 >
+postInitialMessageEvent
+postMessageEvent
+onFirstMessageReceived
+onMessageReceived_
-onObstacleCleared_
-onObstacleTooClose_
-params_
smacc2::client_core_components::CpTopicSubscriber< sensor_msgs::msg::PointCloud2 >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
uint64_t
array< float, 4 >
atomic< float >
atomic< int >
-subscriber_
-lastYawStampUs_
-attitudeQ_
-lastMinRange_
-yawRate_
-lastHits_
cl_px4_mr::CpForwardObstacleGuard
+ CpForwardObstacleGuard()
+ ~CpForwardObstacleGuard()
+ onInitialize()
+ onObstacleTooClose()
+ onObstacleCleared()
+ reset()
+ isTooClose()
+ lastYawRate()
+ lastMinRange()
+ lastHits()
+ params()
- onCloud()
- onAttitude()
- levelRotation()